

TAQHI SHOPPING MALL

AI-Powered Retail Intelligence & Product Safety Operations

A local-first retail operations platform integrating real-time weather signals, SQL product catalogs, and PDF safety guidelines into an explainable, agentic workflow.

- LangChain
- LangGraph
- Ollama (Qwen3 4B)
- FAISS RAG
- FastAPI
- MSSQL
- Streamlit
- LangSmith

3 WEEKS

Build Duration

20 SLIDES

Architecture Deck

5 API

FastAPI Endpoints

100% LOCAL

LLM & Vector Search

01 • STORY & EXECUTIVE SUMMARY

Slides 01 - 03

The Challenge

Build an explainable system beyond a chatbot: connect real data, tools, PDF RAG, weather signals, SQL DB, and a dashboard into a unified workflow.

The Journey

3 weeks of research, design, debugging, and integration. Each iteration refined the platform toward an interview-ready, production-grade architecture.

The Outcome

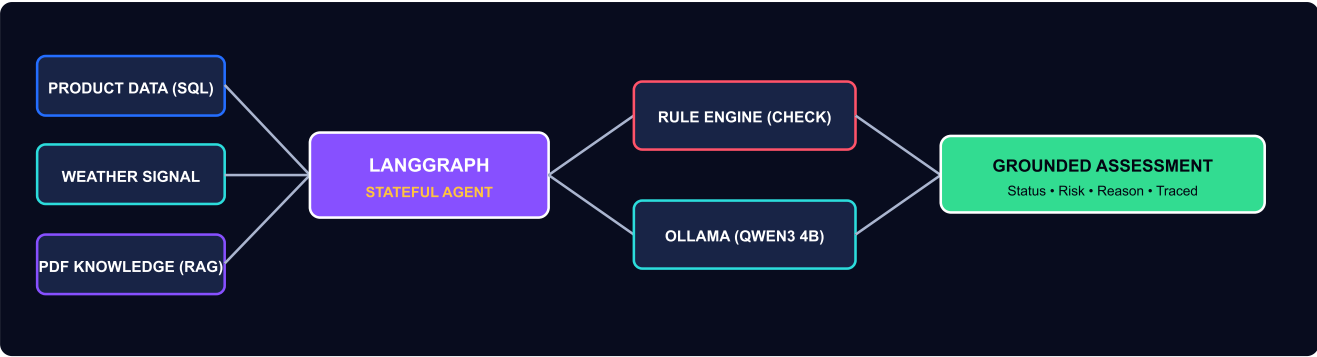
A local-first retail operations platform using Ollama, MSSQL, FAISS, FastAPI, and Streamlit—demonstrating complete end-to-end engineering ownership.

02 • BUSINESS PROBLEM & OPERATIONAL CONTEXT

Slide 03

Retail environments face operational risks when perishable or regulated products meet extreme ambient heat, high humidity, or delayed transport. Operators need instant, grounded answers to evaluate environmental safety compliance without relying on hallucinated AI constraints.

- Dynamic Environmental Signals:** Ambient weather directly impacts storage conditions during transit and floor receiving.
- Expiry & Compliance Enforcement:** Expiry dates must be deterministically checked against real-time system timestamps.
- Grounded Evidence Chain:** System recommendations must reference validated PDF technical specifications and database limits.



04 • LAYERED ARCHITECTURE STACK

Layer	Component / Tool	Core Responsibility
User Experience	Streamlit Dashboard	Interactive operations cockpit with real-time weather & assessments
Service Layer	FastAPI Framework	REST endpoints: /health, /weather, /products, /ask
Orchestration	LangGraph StateGraph	Stateful agent loop coordinating deterministic tools & execution flows
Intelligence	Ollama + Qwen3 4B	Local, zero-cost LLM reasoning & natural language explanation
Knowledge & RAG	FAISS + HuggingFace	Vector search retrieving indexed product safety & SDS PDF chunks
Data Persistence	MSSQL Server + Open-Meteo	Structured product specifications & live ambient weather signals

05 • ENVIRONMENTAL SAFETY & RISK ENGINE LOGIC

Deterministic Safety Rules

The safety logic compares live ambient signals (Temp, Humidity) against SQL threshold parameters. *Pressure checks are executed only if explicitly defined in product specifications.*

Risk Matrix Classifications

Outputs **LOW** (all signals within bounds), **MEDIUM** (approaching limits / upcoming expiry), or **HIGH** (threshold breach or expired product).

06 • ENGINEERING DISCIPLINE, CI/CD & SECURITY

Slides 14, 15 & 16

Observability

Integrated with **LangSmith** for full call tracing, step latency tracking, and tool execution debugging.

Automated CI/CD

GitHub Actions pipeline running **Ruff** linting, **Pytest** unit tests, and automated Docker build image releases.

Security & Safeguards

No hardcoded credentials (env-based configuration) and clear segregation of synthetic test limits from real SDS rules.

07 • 60-SECOND LIVE INTERVIEW DEMO STORYLINE

Slide 18

Step	Action	Demonstrated Capability
01	Open Dashboard	Display real-time weather integration & active product status metrics.
02	Query Product	Submit queries regarding specific SKUs (e.g., GROC-001).
03	SQL Grounding	Show automated retrieval of specific storage limits and expiry dates.
04	RAG Context	Demonstrate FAISS vector search sourcing safety manual PDF excerpts.
05	Risk Evaluation	Display deterministic assessment calculation (Low / Medium / High).
06	Trace Verification	Open LangSmith to show step-by-step agent trajectory and tool execution.

08 • PROJECT SHOWCASE & CONTACT DETAILS

Slide 20

Developer Statement

"Over three weeks of persistent iteration, I engineered **Taqhi_Shopping_Mall** to bridge the gap between autonomous AI agents and deterministic business software. The platform proves how local LLM reasoning can be safely coupled with structured databases and vector knowledge bases."

GitHub Repository

github.com/Taqhi-adf/Taqhi_Shopping_Mall.git

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